

Project Initialization and Planning Phase

Date	23 June 2026
Team ID	SWTID-2026-1377
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	3 Marks

Define Problem Statements:

Traditional assessment methods evaluate students based on keywords or objective questions rather than their actual conceptual understanding. This makes it difficult to identify whether a student truly understands a topic or has misconceptions. The proposed **Voice-Based Concept Understanding Analyser** addresses this challenge by analyzing students' spoken explanations, evaluating their conceptual knowledge using NLP and AI techniques, identifying missing concepts and misconceptions, and providing personalized feedback with a mastery score to support effective learning.

Example:



Problem Statement (PS)	I am (Student)	I'm trying to	But	Because	Which makes me feel
-------------------------------	-----------------------	----------------------	------------	----------------	----------------------------

PS-1	A student preparing for exams or interviews.	Explain a concept using my own voice and evaluate my understanding.	Traditional assessments only check keywords or objective answers and fail to assess true conceptual understanding.	I need detailed feedback that identifies missing concepts, misconceptions, and my level of mastery.	Confident that I can improve my conceptual knowledge and perform better in academics and interviews.
-------------	--	---	--	---	--